25900641 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

MG3DSESDV200 – ACOUSTICS AND PSYCHOACOUSTICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyze (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

ANSWER KEY

Part A – Very Short Answers (2 Marks Each) Answer any 8 questions

1. Typical time gap between direct sound and early reflections that creates intimacy: **10–20 ms**.
2. Room modes: Resonances at specific frequencies in enclosed spaces due to reflec- tions.
3. Convex surfaces scatter sound in many directions, reducing echoes and improving diffusion.
4. Reverberation control ensures clarity and intelligibility of speech and music.
5. 6–16 kHz is critical for vertical localization due to pinna-related spectral cues.
6. Smooth, flat, hard surfaces obey the law of reflection most strongly.
7. Frequency is physical (Hz), pitch is perceptual.
8. Example of strong absorber: **Acoustic foam or mineral wool.**
9. Binaural playback requires **headphones**.
10. Inverse square law: Sound intensity decreases with the square of distance.

Part B – Short Answers (6 Marks Each) Answer any 5 questions

11. Relative distance perception is more accurate since cues are compared (loudness, reverb ratio) rather than judged absolutely.
12. Echo: single delayed reflection (¿50 ms). Flutter echo: multiple rapid reflections between parallel walls.

13. Porous absorbers convert sound energy into heat via air friction; effective for mid-high frequencies.
14. RT60 = 0.161(V/A); T20/T30 used as full 60 dB decay is impractical.
15. Musicians exhibit stronger cocktail party effect due to trained auditory attention.
16. Sabine's formula: RT = 0.161(V/A) = 0.161(3000/300) = **1.61 s.**
17. Typical NC levels: classrooms 30–35, studios 20–25, offices 40–45.

Part C – Essay Questions (12 Marks Each) Answer any 2 questions

18. **Direct Sound, Early Reflections, and Reverberation:**
    - Direct sound provides clarity and localization.
    - Early reflections (10–50 ms) enhance intimacy.
    - Reverberation adds spaciousness and envelopment.
    - Balanced reflections produce natural, immersive sound.
    - Excess reflections cause muddiness; insufficient ones cause dryness.
19. **Applications of SPL Measurement:**
    - Workplace safety: maintain levels below 85 dB.
    - Environmental monitoring: control traffic/industrial noise.
    - System calibration and equalization in audio setups.
    - SPL meters use A-weighting to reflect human sensitivity.
20. **Reverberation in Small vs Large Spaces:**
    - Small rooms: short RT (0.3–0.6 s) for clarity.
    - Large halls: longer RT (1.8–2.5 s) for warmth and blend.
    - Long RT harms speech intelligibility.
    - Short RT reduces musical richness.
    - Multipurpose halls use variable acoustic panels.